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Agentic AI (IA462), Semester VIII

Instructions

1. Save your file as LAB_No_Roll_No.doc (e.g. LAB_5_U21AI001.doc).
2. Paste a snapshot of each step followed along with the result(s).
3. Record your observations and comments after the results.

Experiment 5

Chain-of-Thought Reasoning and Self-Consistency by Sampling

Aim

To generate several sampled reasoning traces for the same question, extract the final answer from each and select the answer by majority vote, comparing accuracy and cost against a single greedy trace and against answering with no reasoning at all.

Theory

A model asked for an answer alone must commit to it in one step. Chain-of-thought prompting asks it to write the intermediate steps first, so the tokens that produce the answer are conditioned on the working rather than on the question alone. The gain is real but fragile: decoding at temperature zero yields exactly one trace, and an error made early in that trace is never revisited, because the model continues to condition on its own mistake.

Self-consistency exploits the fact that there is more than one way to reach a correct answer and many ways to reach a wrong one. The same question is sampled k times at a non-zero temperature; the final answer is extracted from each trace and the most frequent answer is returned. Correct traces tend to agree while errors scatter, so the mode of the answer distribution beats the single greedy trace — which is not the same thing as the most likely answer, since greedy decoding picks the most likely next token at every step, not the most likely conclusion. The price is k times the model calls, measured here alongside accuracy.

The model side is scripted here, as in Experiment 3, so that the machinery is visible without an API key; a real instruction-tuned model changes only the function that returns a trace. Note where the pattern breaks: the vote is taken over extracted answers, so a trace that states its result in an unexpected form contributes nothing, and 2.67 and 2.670 split a vote unless they are normalised first.

Table 1. The three configurations to be compared.

Configuration	Prompt	Decoding	Calls per question	Answer selected by
Direct	question only	temperature 0	1	the single trace
CoT greedy	question with worked examples	temperature 0	1	the single trace
CoT self-consistency	question with worked examples	temperature 0.7	k	majority vote over k traces

Algorithm

1. Build a question set of at least eight problems with their gold answers, write the direct and the chain-of-thought prompt templates, and implement extract_answer(), which finds the stated final answer in a trace, normalises it to a canonical form and returns nothing when no answer can be found.
2. Implement run(configuration): take one trace per question for the two greedy configurations and k sampled traces per question for self-consistency; extract every answer, discard the ones that could not be parsed, return the most frequent remaining answer, and record for each configuration the accuracy, the number of model calls and the number of traces from which no answer could be extracted. Print the vote tally for every question.

Procedure and Result

Paste the snapshots of the steps followed and the output produced by the program below, including the two prompt templates, one complete chain-of-thought trace, the vote tally for every question, and the comparison of the three configurations by accuracy, model calls and unparsable traces.

Observations

Comment on which questions self-consistency corrects that the greedy trace gets wrong and what their vote tallies have in common, on what a narrow winning margin tells you about the answer that won, on the number of model calls spent per additional correct answer, and on why traces from which no answer could be extracted must be counted separately from traces that answered wrongly.

Exercises

1. Sweep k over 1, 3, 5, 9 and 15, repeating each setting with several different seeds, and report accuracy together with total model calls for each k. State the value of k beyond which accuracy stops improving appreciably and the number of calls spent per correct answer at that point.
2. Add an abstention rule: return an answer only when the winning answer leads the runner-up by at least a stated margin, and otherwise report that the system declines to answer. Report coverage and the accuracy on the answered subset for two different margins, and comment on the trade-off.